



BARKER COLLEGE

**TRIAL HIGHER SCHOOL CERTIFICATE
1999**

**MATHEMATICS
3 UNIT (ADDITIONAL)
AND
3/4 UNIT (COMMON)**

BHC
EH
BJR
LJP
GJR
RMH
CLK

PM TUESDAY 17 AUGUST

130 copies

TIME ALLOWED : TWO HOURS
[Plus 5 minutes reading time]

DIRECTIONS TO STUDENTS:

- Write your Barker Student Number on **EACH AND EVERY** page.
- Students are to attempt **ALL** questions.
ALL questions are of equal value. [12 marks]
- The questions are not necessarily arranged in order of difficulty.
Students are advised to read the whole paper carefully at the start of the examination.
- **ALL** necessary working should be shown in every question.
Marks may be deducted for careless or badly arranged work.
- Begin your answer to each question on a **NEW** page. The answers to the questions in this paper are to be returned in **SEVEN SEPARATE BUNDLES**.
Write on **ONLY ONE SIDE** of each page.
- Approved calculators and geometrical instruments may be used.
- A table of Standard Integrals is provided at the end of the paper.

QUESTION 1. (Start a **NEW** page)

Marks

(a) Find $\lim_{x \rightarrow 0} \frac{\sin 5x}{2x}$ 1

(b) Evaluate (i) $\int_0^1 \frac{e^{2x}}{e^{2x} + 1} dx$ 2

(ii) $\int_0^4 \frac{3}{\sqrt{16 - x^2}} dx$ 2

(c) Solve $\frac{2x}{x - 1} > 1$ for all real x . 2

(d) A and B are the points (4, 5) and (8, -1) respectively. 2
Find the point P which divides the interval AB externally in the ratio 3 : 5.

(e) Find the acute angle between the curves $y = \log_e x$ and $y = 1 - x^2$ 3
at the point P (1, 0).

QUESTION 2. (Start a **NEW** page)

Marks

- (a) (i) Write down the expansion of $\cos(\alpha + \beta)$. 3
- (ii) Hence, or otherwise, find the exact value of $\cos 105^\circ$.
- (b) A debating team consists of 12 students, 8 of whom are girls. 3
If three students are chosen at random, what is the probability of selecting
- (i) no girls at all
- (ii) exactly one girl
- (iii) at least two girls ?
- (c) Prove that $\frac{\sin 2\theta}{1 + \cos 2\theta} = \tan \theta$ 2
- (d) Use the substitution $u = 1 - x$ to find the exact value of the integral 4
- $$\int_0^1 x\sqrt{1-x} \, dx$$

QUESTION 3. (Start a NEW page)

Marks

- (a) Melinda invites eleven guests to dinner to celebrate her birthday. 3
Everyone is randomly seated about a round table. Find
- (i) the number of seating arrangements that are possible.
- (ii) the probability that a particular couple, Stuart and Rachael, sit together.
- (b) (i) State the domain and range of the function $f(x) = \cos^{-1} 2x$. 3
- (ii) Draw a neat sketch of the function $f(x) = \cos^{-1} 2x$, clearly labelling all essential features.
- (c) (i) Find the exact value of $\tan^{-1}(\sqrt{3}) - \tan^{-1}(-1)$. 3
- (ii) Hence, or otherwise, find the area bounded by the curve $y = \frac{1}{4 + x^2}$, the x-axis and the ordinates $x = -2$ and $x = 2\sqrt{3}$.
- (c) Prove by Mathematical Induction that $7^n - 1$ is divisible by 6 for all positive integers of n . 3

QUESTION 4. (Start a NEW page)

Marks

- (a) Given that $\sin x > 0$, differentiate $y = \sin^{-1}(\cos x)$, simplifying your answer fully. 2
- (b) Find the term independent of x in the expansion of $\left(x + \frac{1}{2x^2}\right)^6$. 3
- (c) Solve the equation $3\sin x + 4\cos x = 2$ for $0 \leq x \leq 2\pi$. 3
- (d) (i) Given the function $f(x) = x - \sin x - 2$ is a continuous function, 4
determine the nature of any stationary points in the domain $0 \leq x \leq 4\pi$ and
show that this function inflects at $x = n\pi$. (where n is any integer)
- (ii) Hence, or otherwise, draw a neat sketch of the function $f(x) = x - \sin x - 2$
over the domain $0 \leq x \leq 4\pi$.

QUESTION 5: (Start a NEW page)

Marks

- (a) Newton's Law of Cooling can be expressed in the form $\frac{dT}{dt} = -k(T - T_o)$ 5
where T_o is the temperature of the surrounding medium and t is the time and
 k is a constant.

- (i) Verify, by substitution or otherwise, that $T = T_o + Ae^{-kt}$ (where A is a constant)
is the solution to the above differential equation.

- (ii) A body whose temperature is $150^\circ C$ is immersed in a liquid kept at a constant
temperature of $70^\circ C$. In 40 minutes, the temperature of the immersed body
falls to $90^\circ C$. How long altogether will it take for the temperature of the body
to fall to $76^\circ C$?

- (b) The rate $\frac{dV}{dt}$ at which a balloon is pumped up is given by $\frac{dV}{dt} = 1000e^{-2t}$ 7

- (i) Prove that the volume V of air present in the balloon at time t seconds is
given by $V = 500(1 - e^{-2t})$.

- (ii) How many seconds does it take before there is 400 cubic units of air in the balloon ?

- (iii) What is the maximum volume of air which the balloon can hold ?

- (iv) Assuming the balloon is spherical, find the rate at which the radius of the balloon is
increasing when the balloon contains 400 cubic units of air.

QUESTION 6. (Start a **NEW** page)

Marks

- (a) Using the fact that $(1 + x)^{m+n} = (1 + x)^m(1 + x)^n$, show that **3**

$$\binom{m+n}{2} = \binom{m}{2} + \binom{n}{2} + \binom{m}{1}\binom{n}{1}$$

- (b) A particle moves in such a way that its displacement x cm from an origin O at any time t seconds is given by the function $x = \sqrt{3} \cos 3t - \sin 3t$. **4**

(i) Show that the particle is moving in simple harmonic motion.

(ii) Find the period of the motion.

(iii) Find when the particle first passes the origin.

- (c) Rambo is at the top P of a 100 metre vertical cliff PQ. A flat plain extends horizontally from the base Q of the cliff. A Sherman tank is situated somewhere on this plain at point T. Rambo fires a mortar shell from P with an initial velocity of $\frac{190}{\sqrt{3}} \text{ ms}^{-1}$ at an angle of θ to the horizontal and the shell lands on the tank 20 seconds later. **5**

(i) Taking the acceleration due to gravity to be 10 ms^{-2} , show that $\theta = 60^\circ$.

(ii) Find the maximum height above the plain that the mortar shell reaches.

QUESTION 7. (Start a NEW page)

Marks

- (a) P and Q are two points on the parabola $x^2 = 4ay$ with coordinates $(2ap, ap^2)$ and $(2aq, aq^2)$ respectively. The tangents at P and Q meet at T which is situated on the parabola $x^2 = -4ay$. **6**

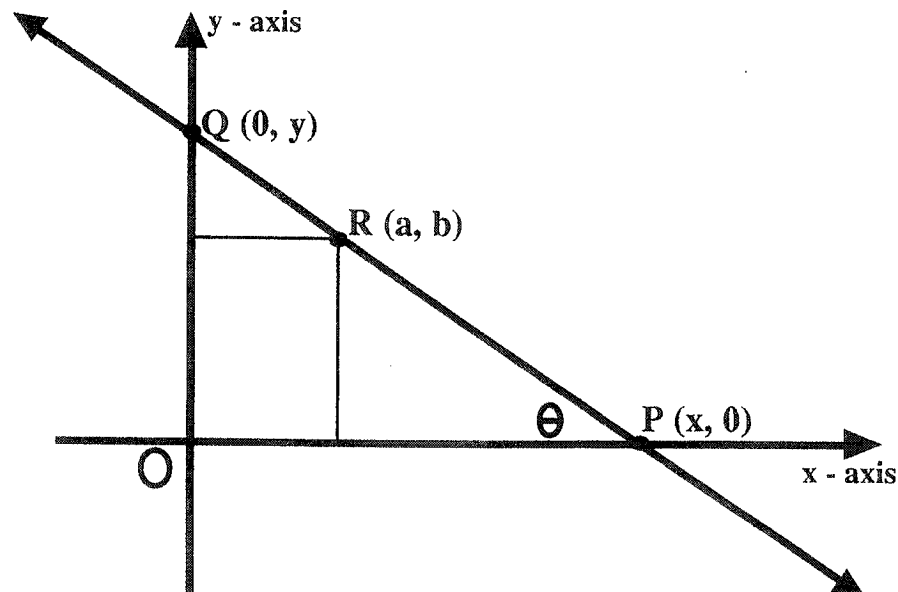
(i) Write down the equations of the tangents at P and Q.

(ii) Show that T is the point $(a(p+q), apq)$.

(iii) Prove that $p^2 + q^2 = -6pq$.

(iv) Find the equation of the locus of the midpoint of PQ.

- (b) The point $R(a, b)$ lies in the positive quadrant of the number plane. **6**
A line through R meets the positive x and y axes at P and Q respectively and makes an angle θ with the x-axis.



(i) Show that the length of PQ is equal to $\frac{a}{\cos \theta} + \frac{b}{\sin \theta}$.

(ii) Hence, show that the minimum length of PQ is equal to $(a^{2/3} + b^{2/3})^{3/2}$.

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 of marks meeting

Question 1

$$\lim_{x \rightarrow 0} \frac{\sin 5x}{2x} = \lim_{x \rightarrow 0} \frac{\sin 5x}{5x} \times \frac{5}{2}$$

$$= 1 \times \frac{5}{2} = \frac{5}{2} \quad (1)$$

$$(i) \int_0^1 \frac{e^{2x}}{e^{2x}+1} dx$$

$$\frac{1}{2} \int_0^1 \frac{2e^{2x}}{e^{2x}+1} dx$$

$$\frac{1}{2} [\ln(e^{2x}+1)]_0^1 \leftarrow (1)$$

$$\frac{1}{2} [\ln(e^2+1) - \ln(e^0+1)]$$

$$\frac{1}{2} [\ln(e^2+1) - \ln(1+1)] \leftarrow (1)$$

$$\frac{1}{2} [\ln(e^2+1) - \ln 2] = \frac{1}{2} \ln\left(\frac{e^2+1}{2}\right)$$

$$\int_0^4 \frac{3}{\sqrt{16-x^2}} dx$$

$$3 \left[\sin^{-1}\left(\frac{x}{4}\right) \right]_0^4 \leftarrow (1)$$

$$\left[\sin^{-1}(1) - \sin^{-1}(0) \right]$$

$$\left(\frac{\pi}{2} - 0 \right)$$

$$\frac{3\pi}{2} \leftarrow (1)$$

$$\frac{2x}{x-1} > 1$$

$$-1 \times \frac{2x}{(x-1)} > 1 \cdot (x-1)^2$$

$$x(x-1) > (x-1)^2$$

$$2x^2 - 2x > x^2 - 2x + 1 \leftarrow (1)$$

$$x^2 - 1 > 0$$

$$(x-1)(x+1) > 0$$

$$x < -1 \text{ or } x > 1 \leftarrow (1)$$

$$\text{Extend } \Rightarrow k: 1 = -3:5$$

$$(e) \text{ For } y = \log_e x, y' = \frac{1}{x}$$

$$\text{When } x=1, m_1 = 1$$

$$\text{For } y = 1-x^2, y' = -2x$$

$$\text{When } x=1, m_2 = -2$$

$$\therefore \tan \theta = \left| \frac{1+2}{1+1(-2)} \right| = \left| \frac{3}{-1} \right| = 3 \quad (1)$$

$$\therefore \theta = 71^\circ 34' \quad (1)$$

Question 2

$$(a) (i) \cos(\alpha+\beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta \quad (1)$$

$$(ii) \cos 105^\circ = \cos(60^\circ + 45^\circ)$$

$$= \cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ \quad (1)$$

$$= \frac{1}{2} \times \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}}$$

$$= \frac{1-\sqrt{3}}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \quad (1)$$

$$= \frac{\sqrt{2}-\sqrt{6}}{4}$$

$$(b) (i) P(\text{no girls}) = \frac{{}^4C_3}{{}^{12}C_3} = \frac{4}{220} = \frac{1}{55} \quad (1)$$

$$(ii) P(\text{exactly 1 girl}) = \frac{{}^8C_1 \times {}^4C_2}{{}^{12}C_3} = \frac{8 \times 6}{220} = \frac{12}{55} \quad (1)$$

$$(iii) P(\text{at least 2 girls}) = 1 - P(\text{No girls or 1 girl})$$

$$= 1 - \left(\frac{1}{55} + \frac{12}{55} \right) \quad (1)$$

$$= \frac{42}{55}$$

$$(c) \text{ LHS} = \frac{\sin 2\theta}{1+\cos 2\theta} = \frac{2\sin \theta \cos \theta}{1+(2\cos^2 \theta - 1)} \quad (1)$$

$$= \frac{2\sin \theta \cos \theta}{2\cos^2 \theta}$$

$$= \frac{\sin \theta}{\cos \theta} \quad (1)$$

$$= \tan \theta = \text{R.H.S}$$

$$\begin{aligned}
 \int_0^1 x\sqrt{1-x} dx &= \int_1^0 -(1-u)\sqrt{u} du \quad (1) \\
 &= \int_1^0 -u^{1/2}(1-u) du \\
 &= \int_1^0 -u^{1/2} + u^{3/2} du \\
 &= \left[-\frac{2u^{3/2}}{3} + \frac{2u^{5/2}}{5} \right]_1^0 \quad (1) \\
 &= 0 + 0 - \left(-\frac{2}{3} + \frac{2}{5} \right) \\
 &= \frac{2}{3} - \frac{2}{5} \\
 &= \frac{4}{15} \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Area} &= \frac{1}{2} \left[\tan^{-1}\left(\frac{x}{2}\right) \right]_{-2}^{2\sqrt{3}} \\
 &= \frac{1}{2} \left[\tan^{-1}(\sqrt{3}) - \tan^{-1}(-1) \right] \\
 &= \frac{1}{2} \times \frac{7\pi}{12} \\
 &= \frac{7\pi}{24} \text{ units}^2 \quad (1)
 \end{aligned}$$

(d) If $n=1$, $7^1-1=6$ which is divisible by 6
 $\therefore$ Statement is true for $n=1$ (1)

Assume statement is true for $n=k$
 i.e. $\frac{7^k-1}{6} = M$ (where M is an integer)

$$\text{i.e. } 7^k - 1 = 6M$$

$$\text{i.e. } 7^k = 6M + 1$$

$$\text{Now, } 7^{k+1} - 1 = 7^k \cdot 7 - 1$$

$$= (6M+1)7 - 1$$

$$= 42M + 7 - 1$$

$$= 42M + 6$$

$$= 6(7M+1) \text{ which is divisible by 6.}$$

$\therefore$ If statement is true for $n=k$, then statement is true for $n=k+1$.

Thus, since statement is true for $n=1$, it is true for $n=2, 3, 4$, etc. (1)

Thus, statement is true for all $n \geq 1$.
 (where n is an integer)

Question 3

i) 12 people

$$\begin{aligned}
 \text{No. of outcomes} &= (12-1)! \\
 &= 11! \quad (1) \\
 &= 39916800
 \end{aligned}$$

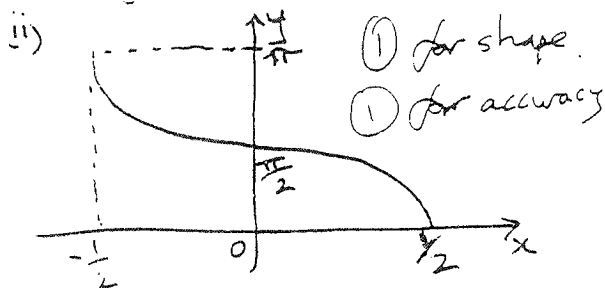
i)  leaving 10 people
 i.e. no. of outcomes = 10!

But can have SR or RS, thus (1) for method
 no. of outcomes = $2 \times 10!$

$$\begin{aligned}
 P(\text{S and R together}) &= \frac{2 \times 10!}{11!} \\
 &= \frac{2}{11} \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 \text{(i) Domain} &= -1 \leq 2x \leq 1 \quad (1) \\
 \therefore -\frac{1}{2} \leq x &\leq \frac{1}{2}
 \end{aligned}$$

$$\text{Range} = 0 \leq y \leq \pi$$



$$\begin{aligned}
 \therefore \text{(i) } \tan^{-1}(\sqrt{3}) - \tan^{-1}(-1) &= \frac{\pi}{3} - \left(-\frac{\pi}{4}\right) \\
 &= \frac{7\pi}{12} \quad (1)
 \end{aligned}$$

Question 4

$$\text{(a) } y = \sin^{-1}(\cos x)$$

$$\therefore \frac{dy}{dx} = \frac{-\sin x}{\sqrt{1-\cos^2 x}} \quad (1)$$

$$= \frac{-\sin x}{\sqrt{\sin^2 x}}$$

$$= \frac{-\sin x}{\sin x}$$

$$= -1 \quad (\text{if } \sin x > 0)$$

$$\begin{aligned}
 \text{(b) General term} &= {}^6C_r x^{6-r} \left(\frac{1}{2x^2}\right)^r \\
 &= {}^6C_r x^{6-r} \left(\frac{1}{2}\right)^r (x^2)^{-r}
 \end{aligned}$$

independent of x occurs when

$$6 - 3r = 0$$

$$\therefore 3r = 6 \text{ is when } r = 2 \quad (1)$$

$$\text{Term} = {}^6C_2 \times \left(\frac{1}{2}\right)^2$$

$$= 15 \times \frac{1}{4}$$

$$= \frac{15}{4} \quad (1)$$

$$3\sin x + 4\cos x = A\sin(x+y)$$

$$= A\sin x \cos y + A\sin y \cos x$$

$$A\cos y = 3 \text{ and } A\sin y = 4$$

$$\cos y = \frac{3}{A} \text{ and } \sin y = \frac{4}{A}$$

$$A = \sqrt{3^2 + 4^2} = 5$$

$$y = \cos^{-1}\left(\frac{3}{5}\right) \div 0.9273$$

$$3\sin x + 4\cos x = 5\sin(x + 0.9273)$$

$$5\sin(x + 0.9273) = 2$$

$$\sin(x + 0.9273) = \frac{2}{5}$$

$$x + 0.9273 \div 0.415, 2.7301, 5.6947$$

$$x \div 1.8028, 5.7674 \quad (1) \text{ for two solutions}$$

$$i) f(x) = x - \sin x - 2$$

$$f'(x) = 1 - \cos x$$

$$f'(x) = 0 \Rightarrow \cos x = 1$$

$$x = \dots, 0, 2\pi, 4\pi, \dots$$

$$y = \dots, -2, 2\pi - 2, 4\pi - 2, \dots$$

x	-1	0	1
y'	+	0	+

$\therefore$ Horizontal pt of inflexion at $(0, -2)$

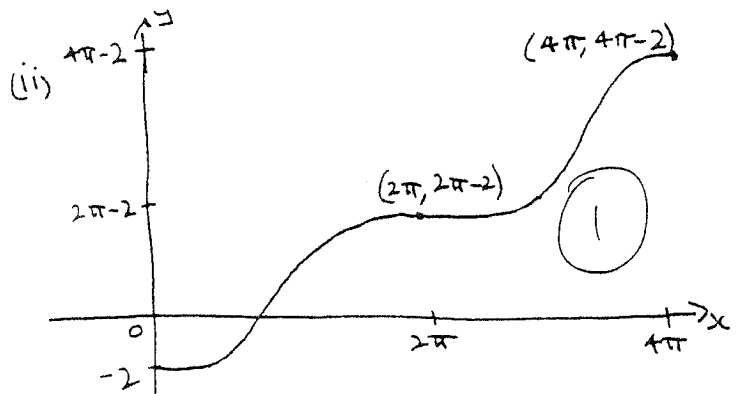
x	6	2π	7
y'	+	0	+

$\therefore$ Horizontal pt of inflexion at $(2\pi, 2\pi - 2)$

x	12	4π	13
y'	+	0	+

$\therefore$ Horizontal pt of inflexion at $(4\pi, 4\pi - 2)$

$$f''(x) = \sin x$$



Question 5

(a) (i) If $T = T_0 + Ae^{-kt}$, then $T - T_0 = Ae^{-kt}$

$$\text{Now, } \frac{dT}{dt} = 0 - kAe^{-kt} = -k(T - T_0)$$

OR using
t-method

① Subst.

② Getting part

③ Answers

$T_0 = 70$ and thus $T = 70 + Ae^{-kt}$

when $t = 0$, $T = 150$

$$\therefore 150 = 70 + Ae^0$$

$$\therefore 150 = 70 + A \therefore A = 80 \quad (1)$$

when $t = 40$, $T = 90$

$$\therefore 90 = 70 + 80e^{-40k}$$

$$\therefore 20 = 80e^{-40k}$$

$$\therefore e^{-40k} = \frac{20}{80} = \frac{1}{4}$$

$$\therefore -40k = \log_e\left(\frac{1}{4}\right)$$

$$\therefore k = \frac{\ln(1/4)}{-40} = \frac{\ln 4}{-40} = \frac{\ln 4}{40} \div 0.0347$$

$$T = 76 \Rightarrow 76 = 70 + 80e^{-kt}$$

$$\therefore 6 = 80e^{-kt}$$

$$\therefore e^{-kt} = \frac{6}{80} = \frac{3}{40}$$

$$\therefore -kt = \log_e\left(\frac{3}{40}\right)$$

$$\therefore t = \frac{\ln(3/40)}{-\ln 4/40} \div 74.74 \text{ minutes}$$

(b) (i) $\frac{dV}{dt} = 1000e^{-2t}$

$$\therefore V = \frac{1000e^{-2t}}{-2} + C$$

$$\therefore V = -500e^{-2t} + C \quad (1)$$

But when $t = 0$, $V = 0$

$$\therefore 0 = -500e^0 + C \therefore C = 500$$

$$\therefore V = 500 - 500e^{-2t}$$

$$e^{-2t} = 1 - \frac{4}{5} = \frac{1}{5}$$

$$-2t = \log_e(1/5) \quad (1)$$

$$t = \frac{\ln(1/5)}{-2} \approx 0.8047 \text{ seconds}$$

$$\text{As } t \rightarrow \infty, e^{-2t} \rightarrow 0$$

$$\therefore (1 - e^{-2t}) \rightarrow 1$$

$$\therefore 500(1 - e^{-2t}) \rightarrow 500 \quad (1)$$

$$\text{Max possible volume} = 500 \text{ units}^3$$

$$\text{Assuming sphere} \Rightarrow V = \frac{4}{3} \pi r^3$$

$$\checkmark \text{ when } V = 400,$$

$$400 = \frac{4}{3} \pi r^3$$

$$\therefore 300 = \pi r^3 \quad (1)$$

$$\therefore r^3 = \frac{300}{\pi} \therefore r = \sqrt[3]{\frac{300}{\pi}} \approx 4.5708$$

$$\text{or, if } V = \frac{4}{3} \pi r^3, \text{ then}$$

$$\frac{dV}{dr} = 4\pi r^2$$

$$\therefore \frac{dr}{dV} = \frac{1}{4\pi r^2}$$

$$\text{at } V = 400, r = \sqrt[3]{\frac{300}{\pi}}, t = \frac{\ln(1/5)}{-2}$$

$$\therefore \frac{dr}{dt} = \frac{dr}{dV} \times \frac{dV}{dt} \quad (1)$$

$$\begin{aligned} \frac{dr}{dt} &= \frac{1}{4\pi r^2} \times 1000e^{-2t} \\ &= \frac{1}{4\pi \left(\sqrt[3]{\frac{300}{\pi}}\right)^2} \times 1000e^{-2 \times \frac{\ln(1/5)}{-2}} \end{aligned}$$

$$\approx 0.7618 \quad (1)$$

Question 6

$$(1+x)^{m+n} = 1 + {}^{m+n}C_1 x + {}^{m+n}C_2 x^2 + \dots$$

$$\text{coefficient of } x^2 = {}^{m+n}C_2 = \frac{(m+n)(m+n-1)}{2} \quad (1)$$

$$(1+x)^m (1+x)^n$$

$$(1 + {}^mC_1 x + {}^mC_2 x^2 + \dots)(1 + {}^nC_1 x + {}^nC_2 x^2 + \dots)$$

containing x^2 will be

$$({}^mC_2 + {}^mC_1 {}^nC_1 + {}^mC_1 {}^nC_2) x^2$$

$\therefore$ Comparing coefficients of x^2 on both sides

$$\binom{m+n}{2} = \binom{m}{2} + \binom{n}{2} + \binom{m}{1}\binom{n}{1}$$

$$(b) (i) x = \sqrt{3} \cos 3t - \sin 3t$$

$$\dot{x} = -3\sqrt{3} \sin 3t - 3 \cos 3t$$

$$\ddot{x} = -9\sqrt{3} \cos 3t + 9 \sin 3t$$

$$= -9(\sqrt{3} \cos 3t - \sin 3t) \quad (1)$$

$$\therefore \ddot{x} = -9x \text{ which is the form } \ddot{x} = -n^2 x$$

$\therefore$ Motion is S.H.M.

$$(ii) \text{ Period} = \frac{2\pi}{3} \quad (1)$$

$$(iii) \text{ when } x = 0,$$

$$0 = \sqrt{3} \cos 3t - \sin 3t \quad (1)$$

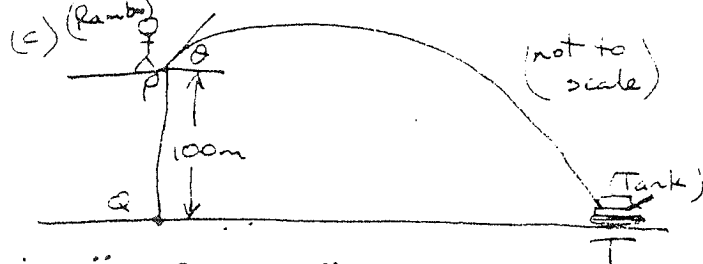
$$\therefore \sin 3t = \sqrt{3} \cos 3t$$

$$\therefore \tan 3t = \sqrt{3}$$

$$\therefore 3t = \frac{\pi}{3}, \frac{4\pi}{3}, \dots$$

$$\therefore t = \frac{\pi}{9}, \frac{4\pi}{9}, \dots \quad (1)$$

$\therefore$ First passes origin at $t = \frac{\pi}{9}$ seconds



$$(i) \ddot{x} = 0 \quad \ddot{y} = -10$$

$$\therefore \dot{x} = C_1 \quad \dot{y} = -10t + C_2$$

$$\text{when } t=0, \dot{x} = V \cos \theta \text{ and } \dot{y} = V \sin \theta$$

$$\therefore \dot{x} = V \cos \theta \quad \dot{y} = -10t + V \sin \theta$$

$$\therefore x = Vt \cos \theta + C_3 \quad y = -5t^2 + Vt \sin \theta + C_4$$

Let P be origin, thus when $t=0$, $x=0$ and $y=0$

$$\therefore C_3 = C_4 = 0$$

$$\therefore x = Vt \cos \theta \text{ and } y = -5t^2 + Vt \sin \theta \quad (1)$$

$$\text{Now, } V = \frac{190}{\sqrt{3}} \text{ and when } t=20, y=-100$$

$$\therefore -100 = -5 \times 20^2 + \frac{190}{\sqrt{3}} \times 20 \sin \theta \quad (1)$$

$$\therefore -100 = -2000 + \frac{3800 \sin \theta}{\sqrt{3}}$$

(iii) Max height occurs when $\dot{y} = 0$

$$\therefore 0 = -10t + \frac{190}{\sqrt{3}} \sin 60^\circ \quad (1)$$

$$\therefore 10t = \frac{190}{\sqrt{3}} \times \frac{\sqrt{3}}{2} = 95$$

$$\therefore t = 9.5 \text{ sec}$$

when $t = 9.5$,

$$y = 100 + (-5 \times 9.5^2 + \frac{190}{\sqrt{3}} \times 9.5 \times \sin 60^\circ)$$

$$y = 100 - 451.25 + 902.5$$

$$\text{Max height} = 551.25 \text{ m} \quad (1)$$

Question 7

$$(i) y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{x}{2a}$$

$$P, m = \frac{2ap}{2a} = p$$

m of tangent at P is

$$y - ap^2 = p(x - 2ap)$$

$$y - ap^2 = px - 2ap^2$$

$$y = px - ap^2$$

$$Q, m = \frac{2aq}{2a} = q$$

m of tangent at Q is

$$y - ag^2 = q(x - 2aq)$$

$$y - ag^2 = qx - 2aq^2$$

$$y = qx - ag^2$$

$$y = px - ap^2 \quad \therefore px - ap^2 = qx - ag^2$$

$$y = qx - ag^2 \quad \therefore px - qx = ap^2 - ag^2$$

$$\therefore x(p - q) = a(p^2 - q^2)$$

$$\therefore x = a(p + q)$$

$$y = ap(p + q) - ap^2$$

$$= ap^2 + apq - ap^2$$

$$= apq$$

$$\therefore = (a(p + q), apq) \quad (1)$$

T lies on parabola $x^2 = -4ay$

$$\therefore (p + q)^2 = -4a^2 pq$$

$$(ii) \text{Midpt of } PQ = \left(\frac{2ap + 2aq}{2}, \frac{ap^2 + ag^2}{2} \right)$$

$$= (a(p + q), \frac{a}{2}(p^2 + q^2))$$

$$\therefore x = a(p + q) \text{ and } y = \frac{a}{2}(p^2 + q^2) \quad (1)$$

$$\therefore \frac{x}{a} = p + q$$

$$y = \frac{a}{2}x - 6pq = -3apq$$

$$\therefore \frac{y}{-3a} = pq$$

Now, if $p^2 + q^2 = -6pq$, then

$$p^2 + 2pq + q^2 = -4pq$$

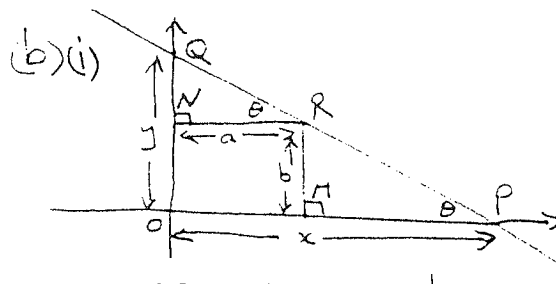
$$\therefore (p + q)^2 = -4pq$$

$$\text{Thus, } \left(\frac{x}{a} \right)^2 = -4 \times \frac{y}{-3a}$$

$$\therefore \frac{x^2}{a^2} = \frac{4y}{3a}$$

$\therefore$ Egn of locus of midpt of PQ is

$$y = \frac{3x^2}{4a} \quad (1)$$



From ΔRPM , $\tan \theta = \frac{b}{x-a}$

$$\therefore \cot \theta = \frac{x-a}{b}$$

$$\therefore b \cot \theta = x - a$$

$$\therefore x = a + b \cot \theta$$

From ΔQRN , $\tan \theta = \frac{y-b}{a}$

$$\therefore a \tan \theta = y - b$$

$$\therefore y = b + a \tan \theta$$

Now, length of PQ = $\sqrt{x^2 + y^2}$

$$\therefore l^2 = x^2 + y^2$$

$$= (a + b \cot \theta)^2 + (b + a \tan \theta)^2$$

$$= a^2 + 2ab \cot \theta + b^2 \cot^2 \theta + b^2 + 2ab \tan \theta + a^2 \tan^2 \theta$$

$$= a^2 + a^2 \tan^2 \theta + b^2 + b^2 \cot^2 \theta + 2ab(\tan \theta + \cot \theta)$$

$$= a^2(1 + \tan^2 \theta) + b^2(1 + \cot^2 \theta) + 2ab \left(\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} \right)$$

(2) for method

$$l^2 = a^2 \sec^2 \theta + b^2 \csc^2 \theta + 2ab \left(\frac{1}{\sin \theta \cos \theta} \right)$$

$$= a^2 \sec^2 \theta + 2ab \sec \theta \csc \theta + b^2 \csc^2 \theta$$

$$l^2 = (a \sec \theta + b \csc \theta)^2$$

$$l = a \sec \theta + b \csc \theta \quad (\text{since } l > 0)$$

$$l = \frac{a}{\cos \theta} + \frac{b}{\sin \theta}$$

$$l = a(\cos \theta)^{-1} + b(\sin \theta)^{-1}$$

$$l' = -a(\cos \theta)^{-2} \times -\sin \theta - b(\sin \theta)^{-2} \times \cos \theta$$

$$= \frac{a \sin \theta}{\cos^2 \theta} - \frac{b \cos \theta}{\sin^2 \theta} \quad \text{--- (1)}$$

$$\text{2. } l' = 0 \Rightarrow \frac{a \sin \theta}{\cos^2 \theta} - \frac{b \cos \theta}{\sin^2 \theta} = 0$$

$$\therefore \frac{a \sin \theta}{\cos^2 \theta} = \frac{b \cos \theta}{\sin^2 \theta}$$

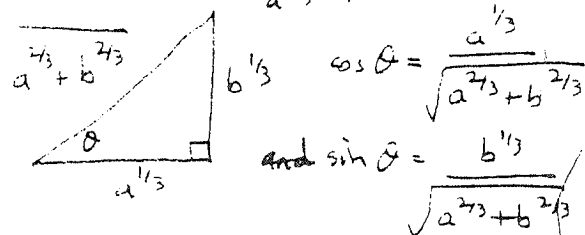
$$\therefore a \sin^3 \theta = b \cos^3 \theta$$

$$\therefore \tan^3 \theta = \frac{b}{a}$$

$$\therefore \tan \theta = \left(\frac{b}{a} \right)^{1/3}$$

$$\therefore \theta = \tan^{-1} \left(\frac{b}{a} \right)^{1/3} \quad \text{--- (1)}$$

$$1. \text{ if } \tan \theta = \frac{b^{1/3}}{a^{1/3}}, \text{ then}$$



$$l = \frac{a}{\frac{a^{1/3}}{\sqrt{a^{2/3} + b^{2/3}}}} + \frac{b}{\frac{b^{1/3}}{\sqrt{a^{2/3} + b^{2/3}}}}$$

$$= a^{2/3} \sqrt{a^{2/3} + b^{2/3}} + b^{2/3} \sqrt{a^{2/3} + b^{2/3}}$$

$$= \sqrt{a^{2/3} + b^{2/3}} (a^{2/3} + b^{2/3})$$

$$= (a^{2/3} + b^{2/3})^{3/2} \quad \text{--- (1)}$$

Alternative way

$$\text{ii) From } \triangle RPM, \cos \theta = \frac{x-a}{PR} \therefore PR = \frac{x-a}{\cos \theta}$$

$$\text{From } \triangle QRN, \sin \theta = \frac{y-b}{QR} \therefore QR = \frac{y-b}{\sin \theta}$$

$$\text{Now } PQ = PR + QR$$

$$\therefore PQ = \frac{x-a}{\cos \theta} + \frac{y-b}{\sin \theta}$$

$$\therefore PQ = \frac{x}{\cos \theta} - \frac{a}{\cos \theta} + \frac{y}{\sin \theta} - \frac{b}{\sin \theta}$$

$$\text{(Now, from } \triangle OPQ, \sin \theta = \frac{y}{PQ} \text{ and } \cos \theta = \frac{x}{PQ})$$

$$\therefore PQ = \frac{y}{\sin \theta} \text{ and } PQ = \frac{x}{\cos \theta}$$

$$\therefore PQ = PQ - \frac{a}{\cos \theta} + PQ - \frac{b}{\sin \theta}$$

$$\therefore PQ = \frac{a}{\cos \theta} + \frac{b}{\sin \theta} \quad \text{--- (2) for Part (i) method}$$

$$\text{OR (ii) } l = \frac{a \sin^3 \theta - b \cos^3 \theta}{\sin^2 \theta \cos^2 \theta}$$

$$= \frac{(a^{1/3} \sin \theta - b^{1/3} \cos \theta)(a^{2/3} \sin^2 \theta + (ab)^{1/3} \sin \theta \cos \theta + b^{2/3} \cos^2 \theta)}{\sin^2 \theta \cos^2 \theta}$$

$$l' = 0 \Rightarrow a^{1/3} \sin \theta - b^{1/3} \cos \theta = 0$$

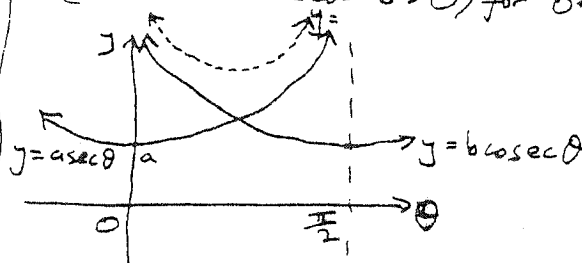
(since $0 < \theta < \frac{\pi}{2}$ and thus

$$(a^{2/3} \sin^2 \theta + (ab)^{1/3} \sin \theta \cos \theta + b^{2/3} \cos^2 \theta) > 0$$

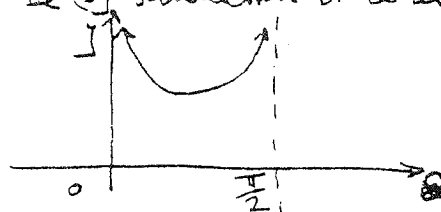
$$\therefore a^{1/3} \sin \theta = b^{1/3} \cos \theta$$

$$\therefore \tan \theta = \frac{b^{1/3}}{a^{1/3}} \therefore \theta = \tan^{-1} \left(\frac{b}{a} \right)^{1/3}$$

To prove minimum value, investigate the graphs of $y = a \sec \theta$ and $y = b \csc \theta$ (where $a > 0$ and $b > 0$) for $0 < \theta < \frac{\pi}{2}$.



thus the graph of $y = a \sec \theta + b \csc \theta$ will be (by summation of ordinates)



$\therefore$ Minimum occurs at $\theta = \tan^{-1} \left(\frac{b}{a} \right)^{1/3}$ where $0 < \theta < \frac{\pi}{2}$